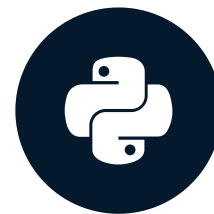


Scope and user-defined functions

INTRODUCTION TO FUNCTIONS IN PYTHON



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Instructor

Crash course on scope in functions

- Not all objects are accessible everywhere in a script
- Scope - part of the program where an object or name may be accessible
 - Global scope - defined in the main body of a script
 - Local scope - defined inside a function
 - Built-in scope - names in the pre-defined built-ins module

Global vs. local scope (1)

```
def square(value):  
    """Returns the square of a number."""  
    new_val = value ** 2  
    return new_val  
  
square(3)
```

9

new_val

```
<hr />-----  
NameError                                Traceback (most recent call last)  
<ipython-input-3-3cc6c6de5c5c> in <module>()  
<hr />-> 1 new_value  
NameError: name 'new_val' is not defined
```

Global vs. local scope (2)

```
new_val = 10

def square(value):
    """Returns the square of a number."""
    new_val = value ** 2
    return new_val

square(3)
```

9

new_val

10

Global vs. local scope (3)

```
new_val = 10

def square(value):
    """Returns the square of a number."""
    new_value2 = new_val ** 2
    return new_value2

square(3)
```

100

```
new_val = 20

square(new_val)
```

400

Global vs. local scope (4)

```
new_val = 10

def square(value):
    """Returns the square of a number."""
    global new_val
    new_val = new_val ** 2
    return new_val

square(3)
```

100

new_val

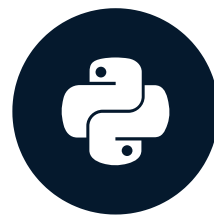
100

Let's practice!

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Nested functions

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Nested functions (1)

```
def outer( ... ):
    """ ... """
    x = ...

    def inner( ... ):
        """ ... """
        y = x ** 2
    return ...
```

Nested functions (2)

```
def mod2plus5(x1, x2, x3):  
    """Returns the remainder plus 5 of three values."""  
  
    new_x1 = x1 % 2 + 5  
    new_x2 = x2 % 2 + 5  
    new_x3 = x3 % 2 + 5  
  
    return (new_x1, new_x2, new_x3)
```

Nested functions (3)

```
def mod2plus5(x1, x2, x3):  
    """Returns the remainder plus 5 of three values."""  
  
    def inner(x):  
        """Returns the remainder plus 5 of a value."""  
        return x % 2 + 5  
  
    return (inner(x1), inner(x2), inner(x3))
```

```
print(mod2plus5(1, 2, 3))
```

```
(6, 5, 6)
```

Returning functions

```
def raise_val(n):  
    """Return the inner function."""  
  
    def inner(x):  
        """Raise x to the power of n."""  
        raised = x ** n  
        return raised  
  
    return inner
```

```
square = raise_val(2)  
cube = raise_val(3)  
print(square(2), cube(4))
```

4 64

Using nonlocal

```
def outer():  
    """Prints the value of n."""  
    n = 1  
  
    def inner():  
        nonlocal n  
        n = 2  
        print(n)  
  
    inner()  
    print(n)
```

```
outer()
```

```
2
```

```
2
```

Scopes searched

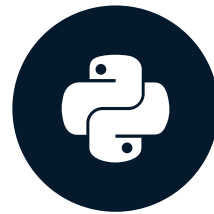
- Local scope
- Enclosing functions
- Global
- Built-in

Let's practice!

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Default and flexible arguments

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You'll learn:

- Writing functions with default arguments
- Using flexible arguments
 - Pass any number of arguments to a functions

Add a default argument

```
def power(number, pow=1):  
    """Raise number to the power of pow."""  
    new_value = number ** pow  
    return new_value
```

```
power(9, 2)
```

```
81
```

```
power(9, 1)
```

```
9
```

```
power(9)
```

```
9
```

Flexible arguments: *args (1)

```
def add_all(*args):  
    """Sum all values in *args together."""  
  
    # Initialize sum  
    sum_all = 0  
  
    # Accumulate the sum  
    for num in args:  
        sum_all += num  
  
    return sum_all
```

Flexible arguments: *args (2)

```
add_all(1)
```

```
1
```

```
add_all(1, 2)
```

```
3
```

```
add_all(5, 10, 15, 20)
```

```
50
```

Flexible arguments: **kwargs

```
print_all(name="Hugo Bowne-Anderson", employer="DataCamp")
```

```
name: Hugo Bowne-Anderson  
employer: DataCamp
```

Flexible arguments: **kwargs

```
def print_all(**kwargs):  
    """Print out key-value pairs in **kwargs."""  
  
    # Print out the key-value pairs  
    for key, value in kwargs.items():  
        print(key + ": " + value)
```

```
print_all(name="dumbledore", job="headmaster")
```

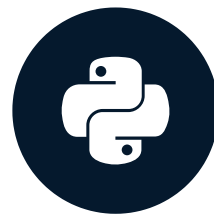
```
job: headmaster  
name: dumbledore
```

Let's practice!

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Bringing it all together

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Next exercises:

- Generalized functions:
 - Count occurrences for any column
 - Count occurrences for an arbitrary number of columns

Add a default argument

```
def power(number, pow=1):  
    """Raise number to the power of pow."""  
    new_value = number ** pow  
    return new_value
```

```
power(9, 2)
```

```
81
```

```
power(9)
```

```
9
```

Flexible arguments: *args (1)

```
def add_all(*args):  
    """Sum all values in *args together."""  
  
    # Initialize sum  
    sum_all = 0  
  
    # Accumulate the sum  
    for num in args:  
        sum_all = sum_all + num  
  
    return sum_all
```

Let's practice!

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